3 URBAN CLIMATE

There is a difference between the wind characteristics outside cities, where big fields of more or less uniform roughness conditions exist, and inside cities, where obstacles and roughness change within short distances (see Figure 18). The wind in cities is less stable and predictable, but turbulent.

Tools like WAsP (Wind Atlas Analysis and Application Program) [xxviii] can not be applied in the city environment. WAsP is a tool to determine wind potential based on information about wind measurements close to the area of interest, roughness parameters of the surroundings and sparsely distributed obstacles in some distance to the point of interest. Usually such tools as WAsP are used to estimate the wind potential of an open and not complex landscape.

To know more about the micro climate on chosen sites in a city environment, possibilities are computational fluid dynamics calculations, on-site measurements, or wind tunnel tests, including the neighbouring area to a sufficient extent. The local wind climate is dependent on its surroundings, so e. g. its distance to the suburbs, to a park, lake or to a high trafficked street. In general the wind velocities in cities are lower and more turbulent.

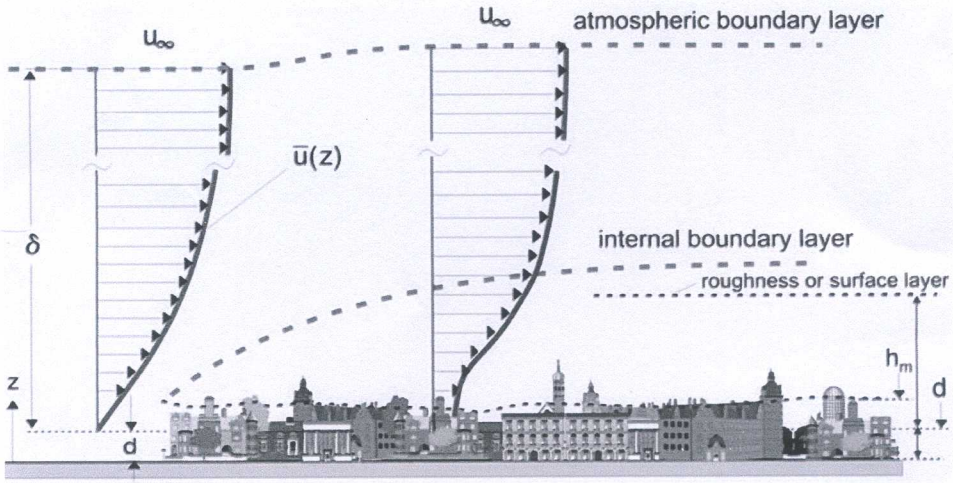

Figure 18: Step change in surface roughness incorporates a changing boundary layer profile [xvii]

3.1 Test Facilities

To simulate the micro wind at the spotted location in a wind tunnel, one possibility is to scale down the city, respect certain roughness and equality rules and chose a method to visualize and/or preferably measure the flow situation. First of all, the boundary layer profile upstream the model shall be shaped. That can happen with the help of vortex generators and roughness elements (see Figure 19). Behind the shaping section follows a measure section to validate the right arrangement of the roughness elements etc. If the model set up and the full-scale conditions shall correspond, equality criteria for the model itself have to be satisfied but also for the incoming flow. Gromke [xxix] investigated the development of the so called Prandtl layer, reaching up to 50-100m in nature and is therefore relevant for urban buildings, city canyons and urban wind energy.

The entire boundary layers can be described by the logarithmic profile or by the power law,

$$\frac{\bar{u}_z}{\bar{u}_{ref}} = \left(\frac{z}{z_{ref}}\right)^{\alpha},$$

with $\bar{u}_z$ as the mean wind velocity at the corresponding height z, $\bar{u}_{ref}$ is a known velocity at the height z_{ref} and α is describing the shape of the profile. For wind profiles in city centres α=0.38 turned out to be appropriated [xxix]. The most important equality number to be kept is the Reynolds number.

It is defined as $$\mathrm{Re}_o = \frac{u_o l_o}{\nu_o} = \mathrm{Re}_m = \frac{u_m l_m}{\nu_m},$$ $= \frac{10 \cdot 10^2}{15 \cdot 10^{-5}} = \frac{1}{15} \cdot 10^8 \approx 67 \cdot 10^6$

where the indices o stands for original and m for model. The parameter l gives a geometrical characteristically length e. g. the height of a certain building, u is the wind velocity and ν is the viscosity of the flowing fluid. In environmental aerodynamics, the model objects are mostly sharp edged and barely aerodynamically shaped as for airplane wings. In this case a limited Reynolds number can be applied. That means, if a Reynolds number of around 10 000 is exceeded, the inert force are very big and the friction forces are predominant. With that the wake area behind, e. g., a house is not changing significantly with Reynolds numbers increasing beyond Re=10 000.

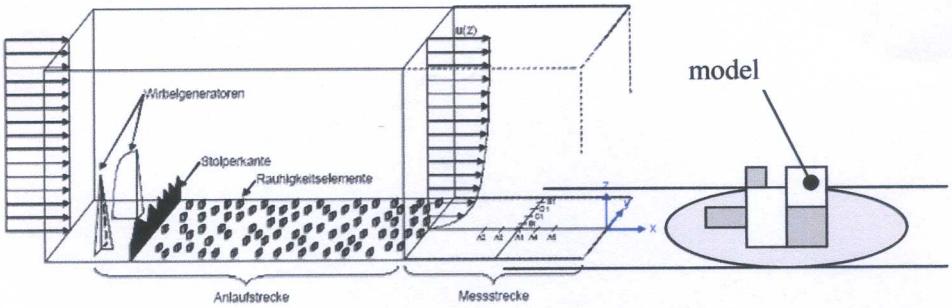

Figure 19: Set up to generate an atmospheric boundary layer in a wind tunnel [xxix]

Considering the height of a tower, l_o=100m, the air viscosity, ν_o=1.5 * 10^{-5} m^2/s, and varying wind velocities with $u_{o,1}$=4m/s (low winds), $u_{o,2}$=5m/s (average wind at ca. 15m without obstacles), $u_{o,3}$=10.3m/s (average wind at 100m), the Reynolds numbers would be Re_1=26.6*10^6, Re_2=33.3*10^6 and Re_3 =68.7*10^6. With these numbers the limit for the restricted Reynolds number is far exceeded. For a wind tunnel model a Reynolds number of at least 10 000 should then be applied. Scaling down the buildings with a factor of 1:300 would result in a tower height of 0.33m and states a size which is feasible in order to fit in a wind tunnel test section. The geometrical scaling factor 1:300 is also satisfying the Reynolds Roughness Criterion for urban districts [xxix]. With the height for the model tower, l_m=0.33m, the same air viscosity and the minimal restricted Reynolds number of Re=10 000, a minimum velocity of u_m=0.4545m/s has to be applied in a wind tunnel. Depending on the measurement device and the purpose of the measurement device a scale of 1:300 might result in too small models. Measurements probably can not be carried out in an accurate way.

3.2 Urban Canyon

Cities are a kind of a landscape. Some areas are lower built, some areas are dominated by very high buildings, in one area there are a lot of parks in the other area there are a lot of parking lots and yet in other areas there is a mix. One very characteristic arrangement is houses aligned along streets. It creates an urban canyon. A lot of investigations were done in this field, mostly concerning pollution or poison gas particle distribution. In case of a perpendicular approach of the wind towards the street, a vortex is developing in the street canyon (see Figure 20, top). Figure 20, middle, depicts the three dimensional vortex situation respecting the bordering buildings or the ends of a long building. If the wind is approaching the street canyon parallel, the wind gets funnelled by the buildings, with that accelerated and behaving like a fluid in a tube (see Figure 20, bottom).

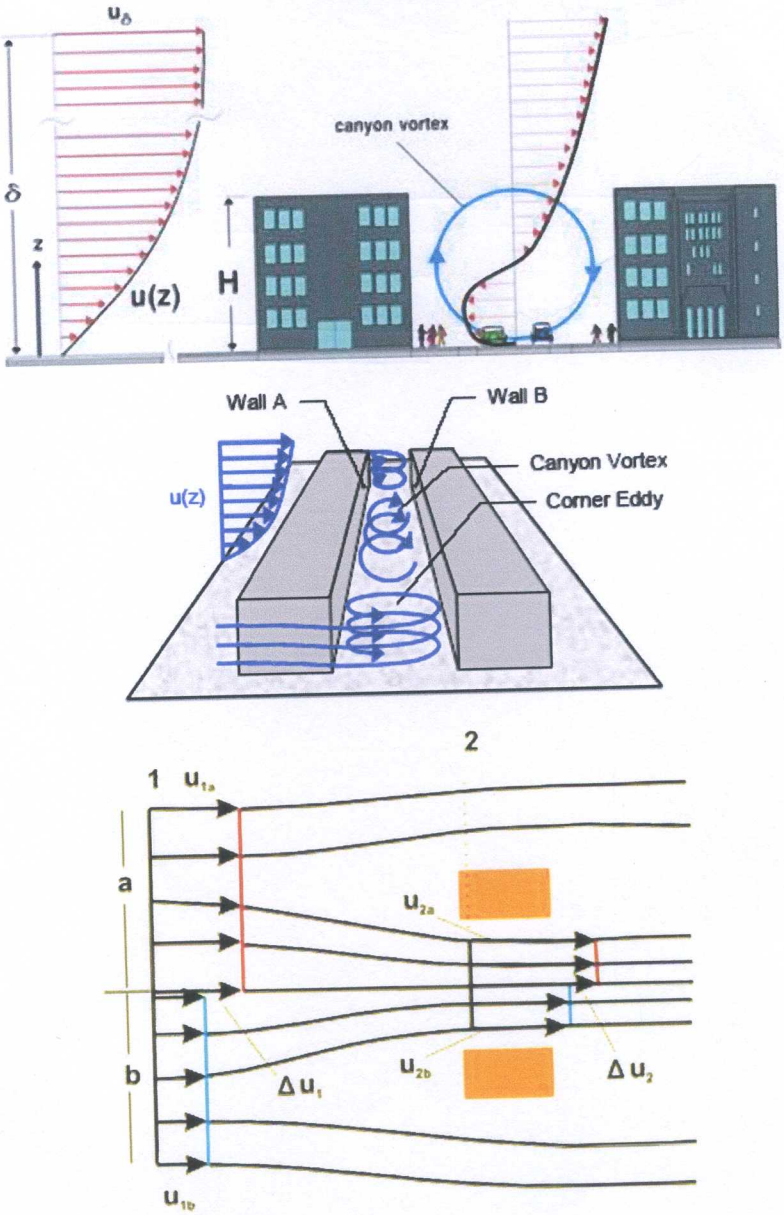

Figure 20: Top: circulation for rectangular approaching winds [xvii]; middle: 3D circulation [xxx]; bottom: funnelling of parallel approaching winds [xvii]